PROVAL co.,Ltd

Calculation Sheet

Calculation header

Identifier	*New calculation (1)*

Medium selection and state

Medium	*Water*
State	*Liquid*

Operating data

☐ Safety-related application

	Maximum flow	Mean flow	Minimum flow	
t1	*83.0*	*46.0*	☒ *0.0*	°C
p1	*8.0*	*8.0*	*0.0*	$kgf/cm^2(g)$
○ p2	*7.0*	*7.0*		$kgf/cm^2(g)$
◉ Δp	*1.0*	*1.0*	☒ *0.0*	kgf/cm^2
Cv	*13.799*	*13.94*	☒	GPM(US)
○ qm	*11,643.0*	*11,882.0*		kg/h
◉ qv	*12.0*	*12.0*	☒ *0.0*	m^3/h
LpAe	*46.3*	*46.0*		dB(A)
	Non-critical	*Non-critical*		

Fluid operating data

	Maximum flow	Mean flow	Minimum flow	
ϱ1	*970.25*	*990.13*	☒	kg/m^3
ϱ2	*970.2*	*990.08*		kg/m^3
○ η1	*0.34169*	*0.58577*	☒	mPa s
◉ ν1	*0.35216*	*0.5916*		mm^2/s
pv1	*0.53476*	*0.10099*	☒	bar(a)
tv1	*174.68*	*174.68*	*99.974*	°C
cF1	*1,555.2*	*1,539.2*		m/s

Pipe downstream of valve

Size class downstream of valve	NPS2	*3/4"*
Schedule downstream	SCH2	*STD*

Valve configuration

Valve type		*Straight globe valve*
Trim type		*Contoured plug*
Flow direction		*FTC*
Valve performance class		*Multi stage valve*
Protection		*Non-hardened*
Low-noise design		*n.a.*

Valve data

Basic characteristic		*Linear*	
☐ Lift stopper			
Selected valve size	NPS	*3/4"*	
Pressure class	class	*class 900*	
Maximum operating pressure	p,max	*157.0*	bar(a)
Valve modifier (Cv/Cv100 = 0.75)	FL^2	*0.892*	-
Valve modifier (Cv/Cv100 = 0.75)	Kc	*0.842*	-
Cavitation modifier (Cv/Cv100 = 0.75)	xFz	*0.632*	-
Valve style modifier (Cv/Cv100 = 1)	Fd	*0.41*	-

Load-dependent values

	Maximum flow	**Mean flow**	**Minimum flow**	
u2	*11.696*	*11.696*		m/s
LpAe	*46.3*	*46.0*		dB(A)
Ri	*0.29*	*0.29*		-

Noise calculation

Calculation standard (noise, liquid)	*IEC 60534-8-4 (2015)*

Noise prediction data

Low-noise design		*n.a.*	
Sound velocity	cF1	*1,555.2*	m/s
Pressure level	LA1	*149.0*	dB
Sound pressure level	Lpi	*149.0*	dB
Transmission loss	TL	*-87.8*	dB

Noise prediction data (continued)

Sound power level (A-weighted)	LwAe	*55.6*	dB(A)
Sound pressure level of valve (A-weighted)	LpAe	*46.3*	dB(A)

Comments:

Reliability index - Ri

High velocity: 100 %
Recommended solution: Standard valve

Reliability index - Ri,op2

High velocity: 100 %
Recommended solution: Standard valve

Legend

Errors